CNENGR202 Instrumentation and Control

Part 2 – L6 RCIM Sensor Characteristics

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Transfer function

- The relationship between its input and its output, and is often expressed by an equation, a graph, or a table

Sensitivity -1

- Defined as the ratio of a small change in the sensor's output to a small change in its input

Parameters of a Pressure Sensor

Input pressure(kPa)		Output voltage(V)	
$P_{in,min}$	40	$V_{out,min}$	1
$P_{in,max}$	90	$V_{out,max}$	5

What is the sensitivity of the pressure sensor?

Sensitivity -2

Input pressure(kPa)		Output voltage(V)	
$P_{in,min}$	40	$V_{out,min}$	1
$P_{in,max}$	90	$V_{out,max}$	5

$$sensitivity = \frac{V_{out,\max} - V_{out,\min}}{P_{in,\max} - P_{in,\min}} = \frac{5-1}{90-40} = 0.08\text{V} \cdot \text{kPa}^{-1}$$

Example

- Two capacitive sensors have sensitivities of 1V/0.1 mm and 1V/0.05 mm, respectively. Which sensor is more sensitive?

Solution

- Two capacitive sensors have sensitivities of 1V/0.1 mm and 1V/0.05 mm, respectively. Which sensor is more sensitive?

- In order to produce a 1V output change, the first sensor must experience a 0.1mm gap change between the capacitor's two plates, while the second sensor only needs a 0.05 mm gap change. Thus, the second sensor is more sensitive

Example-2

Two resistive blood pressure sensors are available: one has a sensitivity of 10mV/V/mmHg; the other has a 7.5 mV/V/mmHg sensitivity. Which sensor should be selected for an application that requires a higher sensitivity?

Solution

- Two resistive blood pressure sensors are available: one has a sensitivity of 10mV/V/mmHg; the other has a 7.5 mV/V/mmHg sensitivity. Which sensor should be selected for an application that requires a higher sensitivity?

- The first sensor produces 10 mV output for each volt of excitation potential and each mmHg of the input pressure, which is higher than 7.5mV output from the second sensor. Thus, the first sensor should be selected.

Sensitivity error or sensitivity drift

- The actual sensitivity deviation from the ideal sensitivity, usually expressed as a percentage

$$\text{Sensitivity error or drift} = \frac{\text{actual sensitivity} - \text{ideal sensitivity}}{\text{ideal sensistitity}} \times 100\%$$

- Such as temperature drift

Example

A resistive blood pressure sensor has an actual sensitivity of 9.5 mV/V/mmHg. Its ideal value is 10 mV/V/mmHg. What is its sensitivity error or drift?

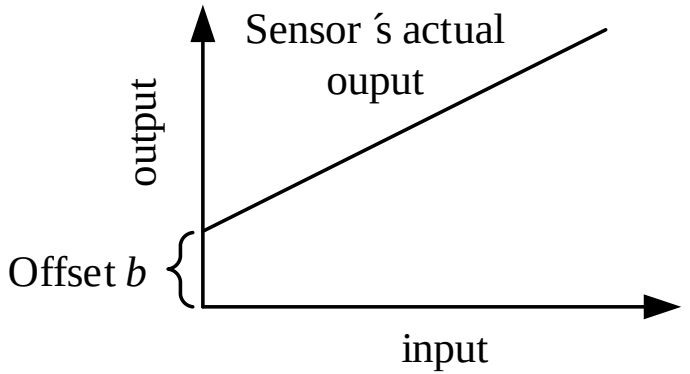
Solution

A resistive blood pressure sensor has an actual sensitivity of 9.5 mV/V/mmHg. Its ideal value is 10 mV/V/mmHg. What is its sensitivity error or drift?

$$\frac{9.5-10}{10}\times 100\% = -5\%$$

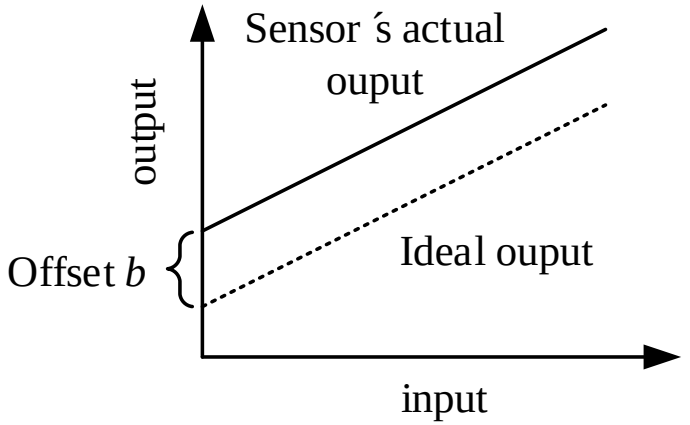
Offset

- When the measured property is zero but a sensor's output value is not zero, then the sensor has an *offset*



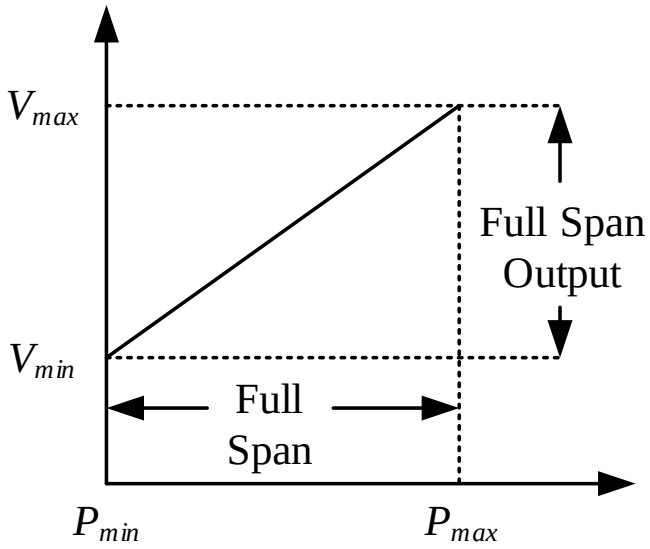
- An offset can be described by either a sensor's output value at its zero input

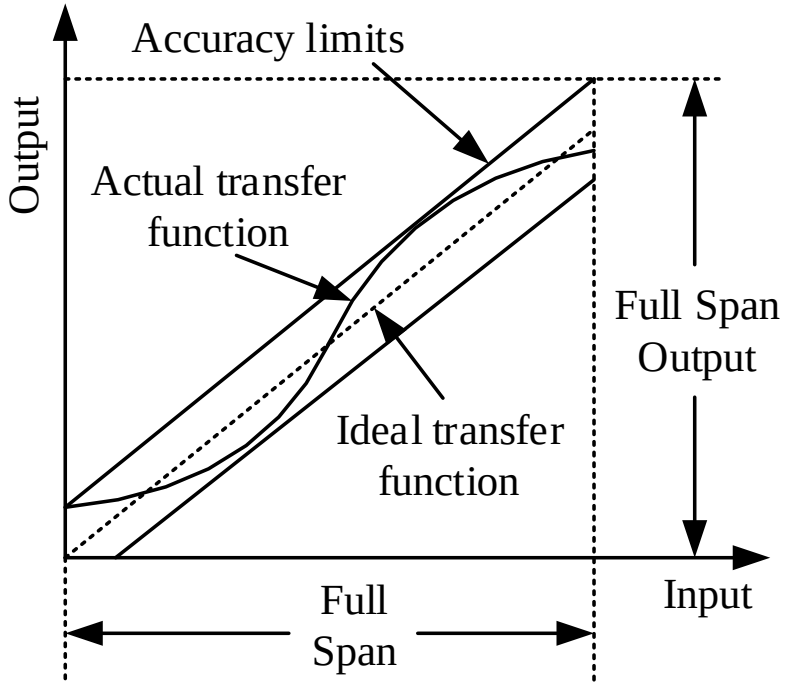
- The difference between the sensor's actual output value and a specified ideal output value



Full span and full-span output

- Full span or full scale (FS): Describe a sensor's measurement range or limitation
- Full-span output (FSO): describe a sensor's dynamic range or output range





Accuracy

- Defined as the maximum deviation of a value measured by the sensor from its true value
- Accuracy can be represented in one of the following forms:
  - *Absolute accuracy*: difference between the measured value and the true value
  - *Relative accuracy*: a percentage of the maximum measurement error versus the true value

$$\frac{\max\left(\text{measured value} - \text{true value}\right)}{\text{true value}} \times 100\%$$

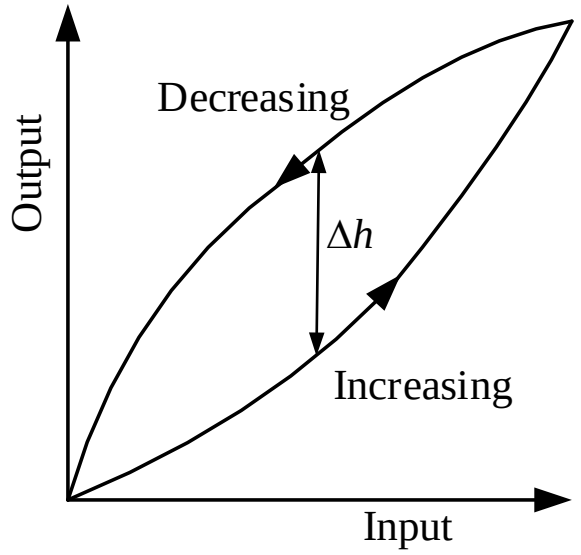
or

$$\frac{\max\left(\text{measured value} - \text{true value}\right)}{\text{full span}} \times 100\%$$

Hysteresis

- A sensor may produce different outputs when measuring the same quantity depending on the “direction” in which the value has been approached
- The maximum difference in terms of the measured quantity is defined as *hysteresis error*

$$\delta_h = \frac{\mathrm{Max}(\Delta h)}{\mathrm{FS}} \times 100\%$$



Example

- A potentiometric displacement sensor produces a maximum of 20 mV difference when measuring the distance of an object passing a certain point: one measurement is taken when the object moves from the left to the right to the point, and the other is taken when the object moves from the right to the left to the point. Find the hysteresis error if the sensitivity of the sensor is 10 mV/mm

Solution

- A potentiometric displacement sensor produces a maximum of 20 mV difference when measuring the distance of an object passing a certain point: one measurement is taken when the object moves from the left to the right to the point, and the other is taken when the object moves from the right to the left to the point. Find the hysteresis error if the sensitivity of the sensor is 10 mV/mm

- Since the hysteresis error is defined as the maximum deviation in terms of the measured quantity, which is displacement in this case, we need to convert the 20 mV output into the displacement input in millimeters:

$$\text{Hysteresis error} = \frac{\text{maximum output voltage difference}}{\text{sensitivity}} = \frac{20\text{mV}}{10\text{mV} \cdot \text{mm}^{-1}} = 2\text{mm}$$

Nonlinearity

- Describes the ‘straightness’ of a sensor’s transfer function.

- A nonlinearity can be expressed by the maximum deviation of a real transfer function from its best-fit straight line, Δl, or in percentage of Δl over the full span:

$$\text{Nonlinearity}(\%) = \frac{\max(\Delta l)}{\text{FS}} \times 100\%$$

Example

- A capacitive displacement sensor's linearity error at its worst point is 0.01 mm. If the full-scale range of its transfer function is 1 mm, what is its nonlinearity expressed as a percentage?

Solution

- A capacitive displacement sensor's linearity error at its worst point is 0.01 mm. If the full-scale range of its transfer function is 1 mm, what is its nonlinearity expressed as a percentage?

$$\text{Nonlinearity}(\%) = \frac{\max(\Delta l)}{\text{FS}} \times 100\% = \frac{0.01}{1} \times 100\% = 1\%$$

Noise

- All sensors generate noise in addition to their output signals
- Sensor noise can be defined as any deviation or fluctuation from an expected value
- A simple average is meaningless since the average of the random variations is zero
- The root mean square (RMS) of the deviations from $V_{average}$ over a time interval T is used:

$$\sqrt{V_n^2} = \sqrt{\frac{1}{T}\int_0^T \left(V - V_{average}\right)^2 dt}$$

V_n^2 : is the mean-square noise in voltage

$\sqrt{V_n^2}$: is the RMS noise in voltage

Signal-to-noise ratio-SNR

- The relative amplitude of the signal of interest to the unwanted noise value

$$SNR = \frac{\overline{P_s}}{\overline{P_n}} = \frac{V_s^2}{V_n^2}$$

Or in decibel (dB)

$$SNR = 10\log_{10}\left(\frac{V_s^2}{V_n^2}\right) = 20\log_{10}\left(\frac{V_s}{V_n}\right)$$

- $\overline{P_s}$: is the average power of the signal
- $\overline{P_n}$: is the average power of the noise
- V_s: is the RMS voltages of the signal
- V_n: is the RMS voltages of the noise

Example-1

- The expected output voltage of a sensor is 100 mV. However, a noise voltage of 18 μV is superimposed on its output signal. Find the SNR of the sensor.

Solution

- The expected output voltage of a sensor is 100 mV. However, a noise voltage of 18 μV is superimposed on its output signal. Find the SNR of the sensor.

$$\text{SNR} = \frac{\left(100\times10^{-3}\right)^2}{\left(18\times10^{-6}\right)^2} = 3.09\times10^{7}$$

Or in decibel (dB)

$$\text{SNR} = 20\log_{10}\frac{100\times10^{-3}}{18\times10^{-6}} = 74.9\text{dB}$$

Example-2

- An amplifier in a sensor circuit has a signal voltage level of 3μV and a noise voltage level of 1μV at input. If the gain of the amplifier is 20 and a 5μV of noise is added by the amplifier at the output, determine the SNR in the input and output

Solution

- An amplifier in a sensor circuit has a signal voltage level of 3μV and a noise voltage level of 1μV at input. If the gain of the amplifier is 20 and a 5μV of noise is added by the amplifier at the output, determine the SNR at the input and output

SNR at input $SNR = \frac{3^2}{1^2} = 9 \quad \rightarrow \quad 9.54$

SNR at output $SNR = \frac{(3 \times 20)^2}{(1 \times 20 + 5)^2} = 5.76 \quad \rightarrow \quad 7.60$

Example-3

- A photodetector has an RMS signal current of 20μA, a shot noise current of 2.27pA, and a thermal noise current of 0.42pA. Find the current SNR of the photodetector

Solution

- A photodetector has an RMS signal current of 20μA, a shot noise current of 2.27pA, and a thermal noise current of 0.42pA. Find the current SNR of the photodetector

The total current noise in RMS can be found from

$$i_n = \sqrt{i_{shot}^2 + i_{thermal}^2}$$

$$\text{SNR} = \frac{i_s^2}{\left(\sqrt{i_{shot}^2 + i_{thermal}^2}\right)^2} = \frac{\left(20\times10^{-6}\right)^2}{\left(2.27\times10^{-12}\right)^2 + \left(0.42\times10^{-12}\right)^2} = 7.65\times10^{13}$$

Resolution-1

- Specifies the smallest change of input parameter that can be detected and reflected in the sensor's output
- Resolution can be described either in absolute terms or relative terms (in percentage of FS)
  - For example, a Honeywell's HMC100 magnetic sensor has a resolution of 27μ gauss (microgauss) in terms of an absolute value, meaning that a minimum change of 27μ gauss can be detected by the sensor
  - an Analog Devices' ADT7301 digital temperature sensor has a resolution of 0.03℃/LSB (least significant bit) , meaning that one bit can distinguish 0.03℃ temperature change
  - a SENSIRION's SHT1 humidity sensor has a resolution of 0.05% relative humidity (RH), meaning that the sensor's resolution is 0.05% of its span

Resolution - 2

- The resolution of a sensor must be higher than the accuracy required in the measurement
  - For instance, if a measurement requires an accuracy within 0.02μm, then the resolution of the sensor must be better than 0.02μm

Precision and repeatability error-1

- Precision refers to the degree of *repeatability* or *reproducibility* of a sensor
- If the exactly same value is measured a number of times, an ideal sensor would produce exactly the same output every time. But sensors actually output a range of values distributed in some manner relative to the actual correct value.

$$\text{Precision} = 1 - \left| \frac{x_n - \bar{x}}{\bar{x}} \right|$$

x_n : is the value of the *n*th measurement

$\bar{x}$: is the average of the set of *n* measurements

Example

- An exact pressure of 150 mmHg was applied to a static pressure sensor. Even though the applied pressure never changed, the output values from the sensor over 10 measurements varied (see Table below). Calculate the average output of the sensor and precision of the sixth measurement

Reading number	Sensor output value(mmHg)
1	149
2	151
3	152
4	149
5	150
6	153
7	152
8	149
9	147
10	151

Solution

The average output of the sensor, $\bar{x}$, is equal to the sum of the 10 measurements divided by 10

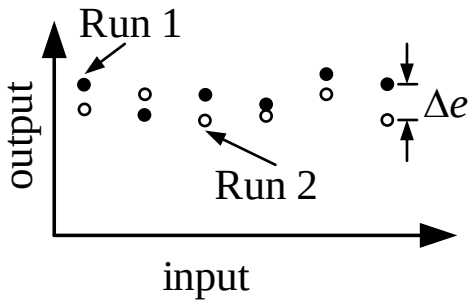
$$\bar{x} = \frac{1}{10}\sum_{n=1}^{10} x_n = \frac{1}{10}\left(149+151+152+149+150+153+152+149+147+151\right) = 150.3$$

The precision of the sixth measurement is

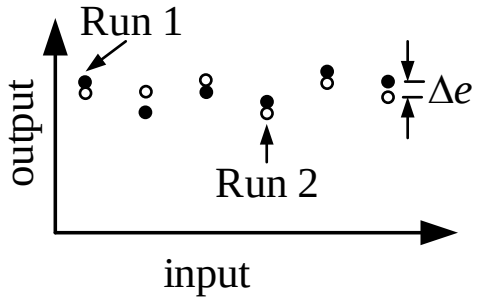
$$\text{Precision} = 1 - \left|\frac{x_6 - \bar{x}}{\bar{x}}\right| = 1 - \left|\frac{153-150.3}{150.3}\right| = 1 - 0.02 = 0.98$$

Precision and repeatability error-2

- *Repeatability error* describes the inability of a sensor to recognize the same value under identical conditions.



Sensor A



Sensor B

- Repeatability error is expressed in a percentage of full scale

$$\delta e = \frac{\max(\Delta e)}{\mathrm{FS}} \times 100\%$$

Calibration and calibration error

- Calibration is to determine the coefficients in their transfer functions (offset, sensitivity)
- Calibration means to convert the sensor's electrical output (e.g., voltage) to a measured value (e.g., temperature)
- The general calibration procedure involves taking a number of measurements over the full operating range and comparing the measured values with actual values (or plotting the sensor's input–output curve against a standard/reference curve), and then adjusting the parameters to match the actual values (or the standard curve)
- The actual values (or the standard curve) may be
  - Obtained using a more accurate sensor as the calibration reference
  - Provided by the sensor's manufacturer
  - Smart or advanced sensors may have self-calibration and adaptive features to adjust themselves to different operating conditions

For Examples

Demonstrate the one-point and two-point calibration procedures of a Part linear temperature sensor. Its transfer function specified by the manufacturer is

$$V_{out} = \left(T_n + 273.15\text{K}\right)\left(10\text{mV/K}\right)$$

V_{out} is its output voltage proportional to the absolute temperature to be measured over its useful range (-40℃ to +100℃) and T_n (in ℃) is the nominal temperature.

Examples-1

- One-point calibration procedures for an LM335 temperature sensor

  One-point calibration is to determine the offset (intercept) of a sensor's transfer function, b, with the assumption that the slope of the transfer function, a, is known (e.g., $a = 1$).

Step 1: Let the temperature being measured by the sensor, T_s, be equal to

$$T_s = aT_n + b = T_n + b \qquad (1)$$

T_n is the nominal value of the temperature being measured, b is the offset before calibration

Step 2: Expose the LM335 sensor to a known and stable temperature, T_a (e.g., T_a= 35℃)

Read and record the temperature measured by the sensor, T_s (e.g., T_s= 34.6℃)

Examples-1

Step 3: Determine the new offset value b' using the formula

$$b' = b + T_a - T_s = b + 35 - 34.6 = b + 0.4$$

Step 4: Replace b with b' in Equation (1)

$$T_s = T_n + b + 0.4$$

Examples-2

- Two-point calibration procedures for an LM335 temperature sensor

  Two-point calibration is to determine both the slope *a* and the offset *b* of a sensor's transfer function

Step 1: Let the temperature being measured by the sensor, T_s, be equal to

$$T_s = aT_n + b \qquad (1)$$

T_n : is the nominal value of the temperature being measured, *a* and *b* are the slop and offset before calibration

Step 2: Expose the LM335 sensor to an accurately known and stable temperature, T_{a1} (e.g.,T_{a1}= 35℃). Read and record the sensor output, T_{s1} (e.g.,T_{s1}= 34.6℃).

Step 3: Expose the LM335 sensor to another accurately known and stable temperature, T_{a2} (e.g.,T_{a2}= 75℃). Read and record the sensor output, T_{s2} (e.g.,T_{s2}= 75.3℃).

Examples-2

Step 4: Determine the new slop value a' and the new offset value b' using the formula

$$a' = ma \qquad b' = n + mb$$

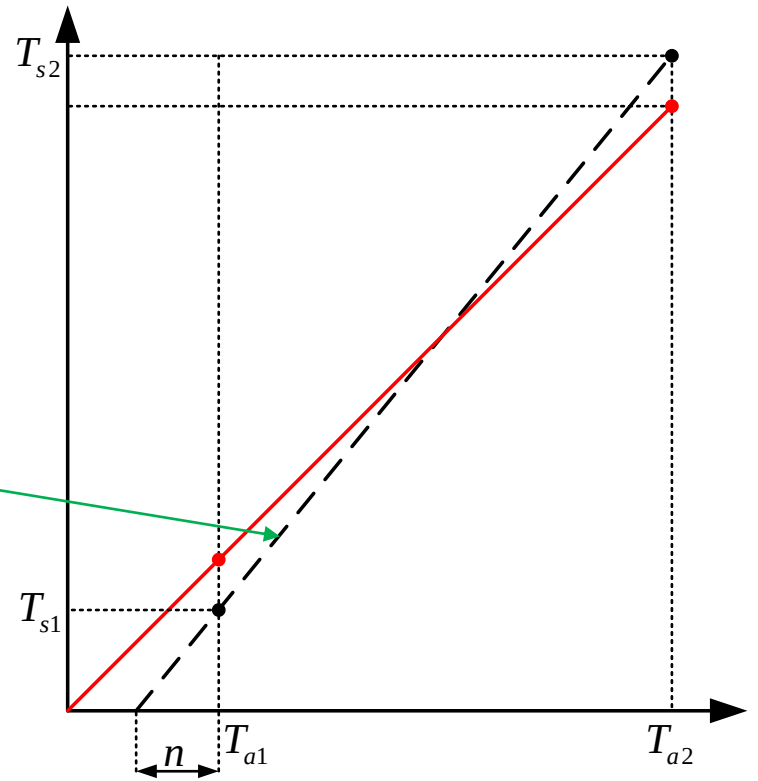
The intermediate values m and n are

$$m = \frac{T_{a1} - T_{a2}}{T_{s1} - T_{s2}} \qquad n = T_{a1} - mT_{s1}$$

$$a' = ma = \frac{T_{a1} - T_{a2}}{T_{s1} - T_{s2}} a = \frac{35 - 75}{34.6 - 75.3} a = 0.98a$$

$$b' = n + mb = \left(T_{a1} - mT_{s1}\right) + mb = T_{a1} + \frac{T_{a1} - T_{a2}}{T_{s1} - T_{s2}}\left(b - T_{s1}\right)$$

$$= 35 + \frac{35 - 75}{34.6 - 75.3}\left(b - 34.6\right) = 1.00 + 0.98b$$



Examples-2

Step 5: Replace a with $a^{'}$ and b with $b^{'}$

$$T_s = (0.98a)T_n + (1.00 + 0.98b) = 0.98(aT_n + b) + 1.00$$

